

PAPER_1678 — Cosmological Flatness $\Omega_k = \Omega_k \sim 1/D_{\text{crit}}^7 = 1.245 \times 10^{-10}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Inflation **Date:** June 18, 2026 **Status:** CLOSED — EXACT formula closure (PAPER_145x-147x broader corpus)

Observation

PAPER_1461 (PAPER_14xx broader corpus, classic cosmology/BSM puzzle) gives:

$$\Omega_k \sim 1/D_{\text{crit}}^7 = 1.245 \times 10^{-10}$$

Status: **EXACT formula.**

UQFF Closed Identity

$$\Omega_k \sim 1/D_{\text{crit}}^7 = 1.245 \times 10^{-10} \quad \text{EXACT formula}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical inflation measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1461
- Related: PAPER_1666-1675 (tier-25); PAPER_1656-1665 (tier-24)
- Calculator dispatch: `calculate_paradox({"paradox": "flatness_1_over_d_crit_7"})`

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